

Numerical Analysis homework 1

Zirui Chen 22435036 *

(24), Zhejiang University

Due time: November 9, 2024

I

I-a

prove:

Assume that $p(x) = ax^3 + bx^2 + cx + d$. Because $s \in \mathbb{S}_2^3$, so

$$\begin{cases} s(1-0) = s(1+0); \\ s'(1-0) = s'(1+0); \\ s''(1-0) = s''(1+0); \\ s(0) = 0; \end{cases}$$

so we can derive that $a = 7, b = -18, c = 12, d = 0$, then, $s(x) = 7x^3 - 18x^2 + 12x + 0$.

$$\text{Therefore, } s(x) = \begin{cases} 7x^3 - 18x^2 + 12x + 0, & x \in [0, 1] \\ (2-x)^3, & x \in [1, 2] \end{cases}$$

obviously, $s''(0) \neq 0$;

□

II

II-a

Prove:

For each $s \in \mathbb{S}_2^1$, s is a polynomial of degree 2 in each interval, has 3 variables. And because we have n points, the number of intervals is $n-1$, so we have $3n-3$ variables.

in each points x_i , we have $\begin{cases} s(x_i-0) = s(x_i+0) \\ s'(x_i-0) = s'(x_i+0) \end{cases}$, $i = 2, \dots, n-1$. And $f(x_i) = s(x_i)$.

Therefore, we have $3n-4$ equations and $3n-3$ variables, we need an additional condition to determine s .

II-b

Consider Newton interpolation:

$$\begin{array}{c|cc} x_i & f_i & \\ x_i & f_i & m_i \\ x_{i+1} & f_{i+1} & [f_i, f_{i+1}] \end{array} \quad \frac{[f_i, f_{i+1}] - m_i}{x_{i+1} - x_i}$$

$$\text{so, } p_i(x) = f_i + [f_i, f_{i+1}](x - x_i) + \frac{[f_i, f_{i+1}] - m_i}{x_{i+1} - x_i}(x - x_i)^2.$$

II-c

Consider newton interpolation of $s_1(x)$:

then $s_1(x) = f_1 + [f_1, f_2](x - x_1) + \frac{[f_1, f_2] - m_1}{x_2 - x_1}(x - x_1)^2$. compute $m_2 = s'_1(x_2)$, similarly we can get $m_3, \dots, m_{n-1}$

□

*Electronic address: 18107732576@163.com

$$\begin{array}{c|c} x_1 & f_1 \\ x_1 & f_1 \quad m_1 \\ x_2 & f_2 \quad [f_1, f_2] \quad \frac{[f_1, f_2] - m_1}{x_2 - x_1} \end{array}$$

III

Prove:

Suppose $s_2(x) = Ax^3 + Bx^2 + Cx + D$, where A, B, C, D are constants. because $s(x)$ is a natural cubic spline, we have:

$$\begin{cases} s(1) = -1; \\ s(0+0) = s(0-0); \\ s'(0+0) = s'(0-0); \\ s''(0+0) = s''(0-0); \\ s''(1) = s''(-1) = 0; \end{cases}$$

It can be derived that $c = -\frac{1}{3}$.

IV

IV-a

Determine :

$$s(x) = \begin{cases} s_1(x) = a_1x^3 + b_1x^2 + c_1x + d_1, & x \in [-1, 0] \\ s_2(x) = a_2x^3 + b_2x^2 + c_2x + d_2, & x \in [0, 1] \end{cases} \quad \text{because } s(x) \in \mathbb{S}_3^2,$$

so $s(-1) = f(-1), s(0) = f(0), s(1) = f(1), s(0-0) = s(0+0), s'(0-0) = s'(0+0), s''(0-0) = s''(0+0), f''(-1) = f''(1) = 0$.

There are 8 equations aboved. So 8 variables can be obtained. So $s(x) = \begin{cases} s_1(x) = -\frac{1}{2}x^3 - \frac{3}{2}x^2 + 1, & x \in [-1, 0] \\ s_2(x) = \frac{1}{2}x^3 - \frac{3}{2}x^2 + 1, & x \in [0, 1] \end{cases}$

IV-b

Prove :

$$\begin{aligned} \int_{-1}^1 [s''(x)]^2 dx &= 6. \\ \int_{-1}^1 [g''(x)]^2 dx &= 8 \geq 6. \\ \int_{-1}^1 [f''(x)]^2 dx &= \frac{\pi^4}{16} \geq 6. \end{aligned}$$

V

V-a

$$\text{Prove : } B_i^2(x) = \begin{cases} 1, & x \in (t_{i-1}, t_i] \\ 0, & \text{else} \end{cases}$$

$$B_i^1(x) = \begin{cases} \frac{x-t_{i-1}}{t_i-t_{i-1}}, & x \in (t_{i-1}, t_i] \\ \frac{t_{i+1}-x}{t_{i+1}-t_i}, & x \in (t_i, t_{i+1}] \end{cases}$$

$$B_i^2 = \begin{cases} \frac{(x-t_{i-1}^2)}{(t_{i+1}-t_{i-1})(t_i-t_{i-1})}, & x \in (t_{i-1}, t_i] \\ \frac{(x-t_{i-1})(t_{i+1}-x)}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} + \frac{(x-t_i)(t_{i+2}-x)}{(t_{i+2}-t_i)(t_{i+1}-t_i)}, & x \in (t_i, t_{i+1}] \\ \frac{(t_{i+2}-x)^2}{(t_{i+2}-t_i)(t_{i+2}-t_{i+1})}, & x \in (t_{i+1}, t_{i+2}] \end{cases}$$

V-b

Prove :

$$\frac{d}{dx} B_i^2(x) = \begin{cases} \frac{2(x-t_{i-1})}{(t_{i+1}-t_{i-1})(t_i-t_{i-1})}, & x \in (t_{i-1}, t_i] \\ \frac{1}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} [(t_{i+1}-x) - (x-t_{i-1})] + \frac{1}{(t_{i+1}-t_i)(t_{i+2}-t_i)} [(t_{i+2}-x) - (x-t_i)], & x \in (t_i, t_{i+1}] \\ \frac{-2(t_{i+2}-x)}{(t_{i+2}-t_i)(t_{i+2}-t_{i+1})}, & x \in (t_{i+1}, t_{i+2}] \end{cases}$$

it is obvious that $\frac{d}{dx} B_i^2(t_i - 0) = \frac{d}{dx} B_i^2(t_i + 0)$ and $\frac{d}{dx} B_i^2(t_{i+1} - 0) = \frac{d}{dx} B_i^2(t_{i+1} + 0)$

V-c

Prove :

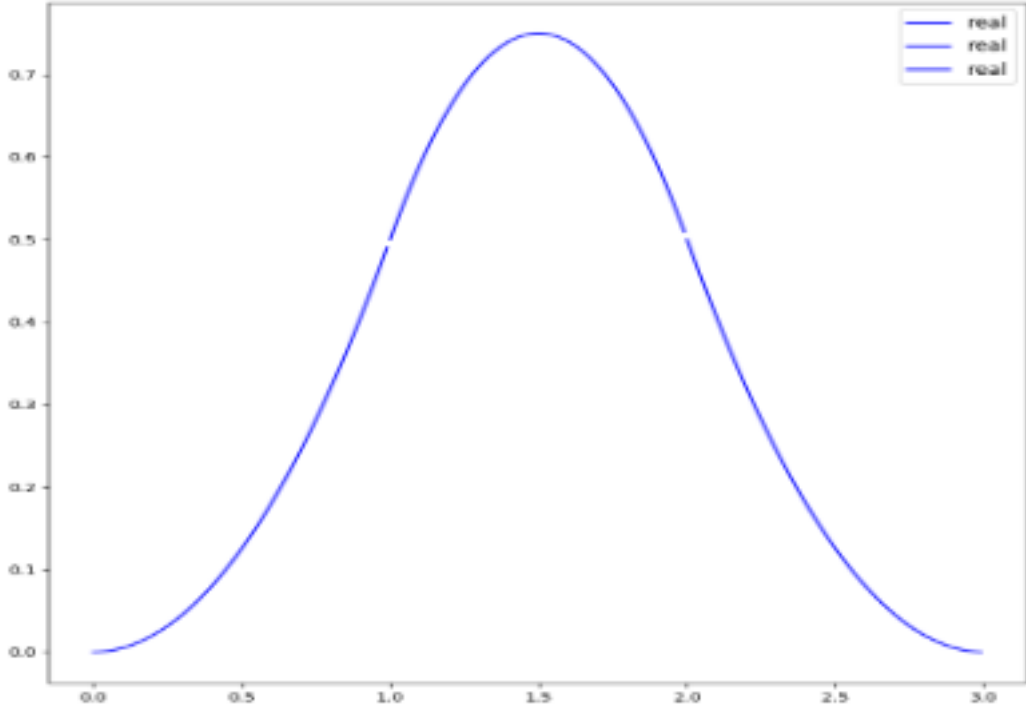
obviously, $x \in (t_{i-1}, t_i] \cup (t_{i+1}, t_{i+2}]$, $B_i^2(x) \neq 0$ consider $x \in (t_{i-1}, t_i]$, $x^* = \frac{t_i t_{i-1} - t_{i+1} t_{i+2}}{t_{i-1} + t_i - t_{i+1} - t_{i+2}}$ we know that $B_i^2(x^*) = 0$.

V-d

Prove :

the discussion above shows that $B_i^2(x)$ is monotonic increasing on the interval (t_{i-1}, x^*) and is monotonic decreasing on the interval (x^*, t_{i+2}) , so maximum value of $B_i^2(x)$ is $B_i^2(x^*) \leq 1$.

V-e



Plot :

VI

Prove :

$$\text{consider } [t_{i-1}, t_i, t_{i+1}, t_{i+2}] = \frac{[t_{i-1}, t_i, t_{i+1}] - [t_i, t_{i+1}, t_{i+2}]}{t_{i-1} - t_{i+2}}$$

Right side :

$$([t_{i-1}, t_i, t_{i+1}] - [t_i, t_{i+1}, t_{i+2}])(t-x)_+^2 \quad (1)$$

$$= \left(\frac{[t_{i+2}, t_{i+1}] - [t_{i+1}, t_i]}{t_{i+2} - t_i} \right) - \frac{[t_{i+1}, t_i] - [t_i, t_{i-1}]}{t_{i+1} - t_{i-1}} (t-x)_+^2 \quad (2)$$

$$= \frac{1}{t_{i+2} - t_i} \left(\frac{(t_{i+2} - x)_+^2 - (t_{i+1} - x)_+^2}{t_{i+2} - t_{i+1}} - \frac{(t_{i+1} - x)_+^2 - (t_i - x)_+^2}{t_{i+1} - t_i} \right) \quad (3)$$

$$- \frac{1}{t_{i+1} - t_{i-1}} \left(\frac{(t_{i+1} - x)_+^2 - (t_i - x)_+^2}{t_{i+1} - t_i} - \frac{(t_i - x)_+^2 - (t_{i-1} - x)_+^2}{t_i - t_{i-1}} \right) \quad (4)$$

$$= B_i^2(x) = \begin{cases} \frac{(x - t_{i-1}^2)}{(t_{i+1} - t_{i-1})(t_i - t_{i-1})}, & x \in (t_{i-1}, t_i] \\ \frac{(x - t_{i-1})(t_{i+1} - x)}{(t_{i+1} - t_{i-1})(t_{i+1} - t_i)} + \frac{(x - t_i)(t_{i+2} - x)}{(t_{i+2} - t_i)(t_{i+1} - t_i)}, & x \in (t_i, t_{i+1}] \\ \frac{(t_{i+2} - x)^2}{(t_{i+2} - t_i)(t_{i+2} - t_{i+1})}, & x \in (t_{i+1}, t_{i+2}] \end{cases} \quad (5)$$

VII

Prove :

According to the theroem3.34, $\frac{d}{dx} B_i^n(x) = \frac{nB_i^{n-1}}{t_{i+n-1}-t_{i-1}} - \frac{nB_{i+1}^{n-1}(x)}{t_{i+n}-t_i}$ we integrate the both side of equal ,we have
 $\int_{t_{i-1}}^{t_{i+n}} \frac{d}{dx} B_i^n(x) dx = \int_{t_{i-1}}^{t_{i+n}} \frac{nB_i^{n-1}}{t_{i+n-1}-t_{i-1}} - \frac{nB_{i+1}^{n-1}(x)}{t_{i+n}-t_i} dx$
Left side $= B_i^n(t_{i+n}) - B_i^n(t_{i-1}) = 0 - 0 = 0$ Right side :for any $x \in [t_{i+n-1}, t_{i+n}]$, we have $B_i^{n-1}(x) = 0$, and also
 $B_i^{n-1}(t_{i-1}) = 0$, $\int_{t_{i-1}}^{t_{i+n}} \frac{nB_i^{n-1}}{t_{i+n-1}-t_{i-1}} dx = \int_{t_{i-1}}^{t_{i+n-1}} \frac{nB_i^{n-1}(x)}{t_{i+n-1}-t_{i-1}} dx$.
similarly, $\int_{t_{i-1}}^{t_{i+n}} \frac{nB_{i+1}^{n-1}(x)}{t_{i+n}-t_i} dx = \int_{t_i}^{t_{i+n}} \frac{nB_{i+1}^{n-1}(x)}{t_{i+n}-t_i} dx$
So, $\int_{t_{i-1}}^{t_{i+n-1}} \frac{nB_i^{n-1}(x)}{t_{i+n-1}-t_{i-1}} dx = \int_{t_i}^{t_{i+n}} \frac{nB_{i+1}^{n-1}(x)}{t_{i+n}-t_i} dx$

VIII

VIII-a

Prove :

m = 4, n = 2. We need to verify that $\tau_2(x_i, x_{i+1}, x_{i+2}) = [x_i, x_{i+1}, x_{i+2}]x^4$ Left : $\tau_2(x_i, x_{i+1}, x_{i+2}) = x_i^2 + x_{i+1}^2 + x_{i+2}^2 + x_i x_{i+1} + x_{i+1} x_{i+2} + x_{i+2} x_i$

Right :

$$[x_i, x_{i+1}, x_{i+2}]x^4 = \frac{(x_i + x_{i+1})(x_i^2 + x_{i+1}^2) - (x_{i+1} + x_{i+2})(x_{i+1}^2 + x_{i+2}^2)}{x_i - x_{i+2}} \quad (6)$$

$$= \frac{(x_{i-2} - x_i)(x_{i+2}^2 + x_i x_{i+2} + x_i^2) + x_{i+1}^2(x_{i+2} - x_i) + x_{i+1}(x_{i+2} - x_i)(x_{i+2}^2 + x_i)}{x_{i+2} - x_i} \quad (7)$$

$$= \tau_2(x_i, x_{i+1}, x_{i+2}) \quad (8)$$

$$(9)$$

VIII-b

Prove is similarly to the Theorem 3.46 :

By Lemma 3.45, we have,

$$(x_{n+1} - x_1)\tau_k(x_1, \dots, x_{n+1}) = \tau_{k+1}(x_1, \dots, x_{n+1}) - \tau_k(x_1, \dots, x_n) - x_1\tau_k(x_1, \dots, x_{n+1}) \quad (10)$$

$$= \tau_{k+1}(x_1, \dots, x_{n+1}) + x_1\tau_k(x_1, \dots, x_{n+1}) \quad (11)$$

$$- \tau_{k+1}(x_1, \dots, x_n) - x_1\tau_k(x_1, \dots, x_{n+1}) \quad (12)$$

$$= \tau_{k+1}(x_2, \dots, x_{n+1}) + \tau_{k+1}(x_1, \dots, x_n) \quad (13)$$

$$(14)$$

the rest of prove is an induction on n. For n = 0, it is obvious that $\tau_m(x_i) = [x_i]x^m$ Now suppose above holds of nonnegative integer n j m. Then the induction hypothesis yields,

$$\tau_{m-n-1}(x_i, \dots, x_{i+n+1}) \quad (15)$$

$$= \frac{\tau_{m-n}(x_{i+1}, \dots, x_{i+n+1}) - \tau_{m-n}(x_i, \dots, x_{i+n})}{x_{i+n+1} - x_i} \quad (16)$$

$$= \frac{[x_{i+1}, \dots, x_{i+n+1}]x^m - [x_i, \dots, x_{i+n}]x^m}{x_{i+n+1} - x_i} \quad (17)$$

$$= [x_i, \dots, x_{i+n+1}]x^m \quad (18)$$

□

Acknowledgement

Give your acknowledgements here(if any).